COURSE TITLE : INJECTION MOULD DESIGN
COURSE CODE : 5094
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52/5
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Various machines used for mould making	16
2	Steps involved in the construction of mould and accessories	12
3	Steps involved in the construction of mould and accessories	12
4	Feed system and cooling in mould	12
TOTAL		**52**

COURSE OUTCOME

Sl.	Sub	Student will be able to
1	1	To understand and the various machines used for mould making
	2	Steps involved in the construction of mould.
3	3	Understand the ejection mechanism Understand ejector plate assembly Explain the ejection techniques
4	4	To understand the feed system and cooling in mould.

SPECIFIC OUTCOME

MODULE I

1.1.0 To understand the various machines used for mould making

1.1.1 List the various machines used
1.1.2 Describe the various parts of lathe
1.1.3 Explain the purpose of lathe
1.1.4 Understand turning, boring and facing
1.1.5 Explain the production of a mould using lathe.
1.1.6 Understand the important parts with sketches, purpose and working of machine tools

1.1.6.1 Shaping machine
1.1.6.2 Planning machine

MODULE II

2.1.0 Steps involved in the construction of mould and accessories

2.1.1 Understand the terms

2.1.1.1 Impression, cavity and core-Sprue bush
2.1.1.2 Runner, gate, register ring, Guide pillars
2.1.1.3 Bushes, fixed half plate, moving half,

2.1.2 Explain the production of cavity and core
2.1.3 Core inserts

2.2.1 Understand the steps involved in Bench fitting

MODULE III

3.1.0 Steps involved in the construction of mould and accessories

3.1.1 Explain the different ejector systems

3.1.1.1 In-line ejector grid system

3.1.2 Understand ejector plate assembly

3.1.2.1 Ejector plate
3.1.2.2 Retaining plate
3.1.2.3 Guiding and supporting ejector plate assembly
3.1.2.4 Ejector rod and ejector rod bush
3.1.2.5 Ejector plate assembly return system

3.2.0 Explain the ejection techniques

3.2.1 Understand the different methods of ejection
3.2.2 Explain the various ejection techniques.

3.2.2.1 Pin ejection
3.2.2.2 Stepped ejection pins
3.2.2.3 D-shaped ejector pin
3.2.2.6 Valve ejection

MODULE IV

Feed system and cooling in mould

4.1.0 To understand the feed system and cooling in mould.

4.1.1 Describe runner in injection mould design

4.1.1.1 Understand the cross-section, runner layout, shape and size of runner
4.1.1.2 Explain the factors affecting runner layout

4.1.2 Define gate.

4.1.3 Explain the different gate
4.1.4 Explain the positioning and balancing of gate
4.1.5 Empirical relationship for gate depth.
4.1.6 Explain location of parting surface.
State the following parting surface
4.1.6.1 Flat parting surface
4.1.6.2 Angled parting surface
4.1.6.3 Stepped parting surface
4.1.6.4 Profiled parting surface
4.1.6.5 Balancing of mould surface

4.2.0 Understand the Methods adopted for mould cooling
4.2.1 Cooling integer type mould plates
4.2.2 Cooling integer type cavity plate
4.2.3 Cooling integer type core plate

4.3.0 Cooling other mould part
4.3.1 Cooling valve-type ejection.
4.3.2 Cooling the the sprue bush-Water connections-adaptors, quick connection adaptors

CONTENT DETAILS

UNIT-I
Machines tools used for mould making

Lathe-important part, Primary purpose, turning, boring and facing-Mould part produced on lathe-Steps involved in the manufacture of a typical mould part-**Cylindrical grinding machine-**important parts with sketches-working-**Shaping machine-**purpose, working- **planing machine-**working –**Surface -** important parts with sketches, purpose, and working of **Milling machine**, **grinding machine and tracer-controlled milling machine**

UNIT-II Mould construction

Basic terminology-Impression, cavity and core-Sprue bush, runner, gate, register ring, Guide pillars& bushes, fixed half plate, moving half,-Manufacture of cavity and core-use of logical inserts, examples-fitting of inserts-multi-impression-mould alignment-Bolsters-different types-Ancillary items-designing of guide pillars,positioning –Attachment of mould platern- Casting-Electrodeposition-Cold hobbing-Pressure casting-Bench fitting

UNIT-III

Ejector systems-ejector grid- different types, in-line, frame-type and block grid-Ejector plate assembly-ejector plate, retaining plate, plate assembly return system, stop pin-Ejection techniques –

different methods, pin ejection, stepped ejection pins, D-shaped ejector pin, sleeve ejection, blade ejection, valve ejection, air ejection, stripper bar ejection, stripper plate ejection- different methods, stripper ring ejection-Ejection from fixed plate-Sprue pullers.

UNIT-IV

Feed System-Runner, cross-section shape, size-Runner layout, factors affecting layout-Gate, requirements-Positioning of gate-Balanced gating-Types of gate-Empirical relationship for gate depth
Parting Surface-flat parting surface, non-flat parting surface-stepped parting surface, profiled parting surface, angled parting surface, complex edge form-Balancing of mould surface- Venting

Mould Cooling-cooling integer type mould plates-cooling integer type cavity plate, cooling integer type core plate-Cooling insert-bolster assembly-cooling bolster, cooling cavity inserts, cooling circular insert, cooling core inserts-Cooling other mould part-Cooling valve-type ejection, cooling the the sprue bush-Water connections-adaptors,quick connection adaptors

REFERENCE

1. Injection Mould design- R.G.W.Pye (Publisher – lliffe 2nd Edition)
2. Plastic Product design-Ronald.D.Beck (Publisher -Van Nostrand Reinhold Co., 1970)
3. Design plastic moulds and dies-LASZLO & IMRE BALZ